CSCI 6461 | TOPIC 2

RISC vs. CISC: Instruction Set Design Principles & Trade-offs

ISA boundaries, microarchitectural costs, and quantitative trade-offs

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Topic 2: Module Roadmap

1. **ISA Foundations & Complexity Distribution**
   - The programmer-visible hardware contract and historical constraints.
2. **The CISC Architectural Model**
   - Register-memory operations, variable-length encodings, and microarchitectural costs.
3. **The RISC Revolution & Principles**
   - Load/store discipline, fixed 32-bit width, and side-by-side assembly comparison.
4. **Quantitative Performance, Energy & Modern Convergence**
   - Iron Law evaluation, Amdahl's Law, performance-per-watt, and modern μop cracking.

PART 1

ISA Foundations & Historical Context

What Is an Instruction Set Architecture?

- An ISA is the **programmer-visible functional specification** of a processor.
- Defines instructions, register files, addressing modes, and architectural state.
- Sits directly at the hardware/software boundary as an abstraction contract.
- **Complexity Redistribution:** ISA design redistributes complexity between software (compilers) and hardware (microarchitectures).

Historical Constraints Shaped Early ISAs (CISC Era)

- **Expensive, Slow Memory (1960s–1970s):** Small core memory and high DRAM latency placed a premium on minimizing binary program size.
- **Assembly-Centric Programming:** Programs were frequently hand-written; rich instructions helped express high-level constructs directly.
- **Microprogrammed ROM Control:** Complex multi-cycle instructions were cheap to implement via on-chip microcode routines.
- **The Semantic Gap:** Early designers believed machine instructions should mirror high-level statements directly.

PART 2

The CISC Architectural Model

The CISC Architectural Model

- **CISC:** *Complex Instruction Set Computer* (e.g., VAX, x86).
- **Core Characteristics:**
  - **Variable-Length Encodings:** Instructions range from 1 to 15 bytes to maximize code density.
  - **Register-Memory Operations:** Arithmetic instructions can accept memory operands directly (e.g., `add [rdi], rax`).
  - **Complex Addressing Modes:** Evaluates scaled indices and indirect pointers in hardware.
- **Microarchitectural Cost:** Requires multi-stage sequential decoders and handles irregular execution latencies.

CISC Execution: Register-Memory Translation

CISC Instruction Example (x86-64)

```
add [rdi + 8], rax
```

Architectural View (Software)

- 1 instruction fetched.
- Compact variable encoding.
- Atomic arithmetic directly to memory.

Hardware Decomposition (μops)

1. Compute effective address (rdi $+ 8$).
2. Load memory contents into ALU.
3. Add contents of register `rax`.
4. Store result back to memory.

One architectural instruction decomposes into multiple hardware micro-operations.

PART 3

The RISC Revolution & Principles

The 1980s Shift: Emergence of RISC

- **Empirical Workload Findings:** Patterson & Hennessy demonstrated that compiled code utilized $< 20\%$ of available CISC instruction repertoires over 90% of the time.
- **Pipeline Hazards:** Variable-length, irregular instructions created severe pipeline stalls and hazard control logic bottlenecks.
- **Compiler Advancements:** Modern optimizing compilers handle register allocation and instruction scheduling more effectively than monolithic hardware.

The RISC Design Principles

- **Fixed 32-Bit Instruction Width:** Simplifies parallel instruction fetch and multi-decoder design.
- **Strict Load/Store Architecture:** ALU operations use *only* registers; memory access is restricted to explicit `ldr`/`str`.
- **Large Orthogonal Register File:** 31 general-purpose registers (AArch64 `x0`–`x30`) maintain active variables on-chip.
- **Single-Cycle Execution Goal:** Uniform pipeline stage latencies prevent pipeline bubbles.

Code Comparison: RISC vs. CISC Memory Addition

High-level statement: `c = a + b;` (operands in memory)

AArch64 RISC Assembly

`ldr x1, [x0, #0]`	load a
`ldr x2, [x0, #8]`	load b
`add x3, x1, x2`	compute sum
`str x3, [x0, #16]`	store c

x86-64 CISC Assembly

`mov rax, [rdi]`	load a
`add rax, [rdi + 8]`	load b + add
`mov [rdi + 16], rax`	store c

RISC uses 4 fixed-width instructions (16 bytes); CISC uses 3 variable-length instructions, but hardware workload executed is identical.

PART 4

Quantitative Performance & Trade-offs

The Performance Equation (Iron Law)

CPU Execution Time Equation

$$T_{\text{CPU}} = IC \times CPI \times T_{\text{clk}} = \frac{IC \times CPI}{f_{\text{clk}}}$$

- **Instruction Count (IC):** Total executed instructions (influenced by ISA and compiler).
- **CPI:** Average clock cycles per instruction (influenced by pipeline and memory stalls).
- **Clock Period (T_{clk}):** Duration of one cycle (influenced by circuit depth and design).

Worked Performance Comparison

Workload evaluation across two competing microarchitectures:

Machine A (RISC / AArch64)

- $IC = 1.4 \times 10^9$
- $CPI = 1.10$
- $f_{\text{clk}} = 3.0$ GHz

$$T_A = \frac{1.4 \times 10^9 \times 1.10}{3.0 \times 10^9} = \mathbf{0.513}\ \text{s}$$

Machine B (CISC / x86-64)

- $IC = 1.0 \times 10^9$ (-28% IC)
- $CPI = 1.60$ (Higher CPI)
- $f_{\text{clk}} = 3.0$ GHz

$$T_B = \frac{1.0 \times 10^9 \times 1.60}{3.0 \times 10^9} = \mathbf{0.533}\ \text{s}$$

RISC wins (Speedup $\approx 1.04\times$) because its lower CPI compensates for executing more total instructions.

Fetch Complexity: Fixed vs. Variable Width

RISC Fixed 32-Bit Decode

- Instruction k is located directly at $\text{pc} + 4k$.
- Enables simple parallel decoders (e.g., 8-wide decode).
- Low front-end power consumption and die area.

CISC Variable-Length Decode

- Lengths span 1 to 15 bytes.
- Requires sequential pre-decoders to determine instruction boundaries.
- High front-end complexity; requires decoded μop caches (DSB).

Amdahl's Law and Performance Limits

Amdahl's Law Equation

$$S_{\text{overall}} = \frac{1}{(1-f) + \frac{f}{S_{\text{enhanced}}}}$$

- f: Fraction of execution time subjected to optimization.
- S_{enhanced}: Speedup factor achieved on that specific fraction.
- **Example ($f = 0.60, S_{\text{enhanced}} = 10$):**

$$S_{\text{overall}} = \frac{1}{0.40 + \frac{0.60}{10}} \approx \mathbf{2.17}\times$$

- **Takeaway:** Overall speedup is strictly capped by the non-accelerated fraction $(1 - f)$.

Energy, Thermal Limits & Performance-per-Watt

Dynamic Power Equation

$$P = C \cdot V_{\text{dd}}^2 \cdot f + P_{\text{static}}$$

- Complex decoders increase switching capacitance (C).
- Higher frequencies (f) require higher voltages (V_{dd}).

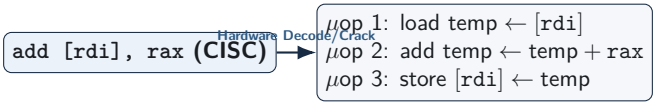
Performance-per-Watt

$$\frac{\text{Workload Throughput}}{\text{Watts}}$$

- Primary metric for mobile batteries and cloud data center racks.
- Streamlined RISC front-ends deliver high IPC at low decode energy.

Microarchitectural Convergence: Modern x86 Internals

Modern x86 processors preserve the CISC ISA externally, but execute a **RISC datapath internally**:



- The execution core is an out-of-order engine processing RISC-like micro-operations (μops).
- CISC-to-RISC translation occurs dynamically, incurring a front-end decode power overhead.

Review and Self-Test Questions

Topic 2: Comprehensive Summary

- **Complexity Redistribution:** ISA design distributes work between compilers and hardware decoders.
- **CISC Trade-offs:** Prioritizes code density and low instruction count via variable-length instructions and register-memory operations, at the cost of front-end decode complexity.
- **RISC Trade-offs:** Prioritizes pipeline regularity, fixed 32-bit width, and explicit load/store architecture to achieve high clock frequency and low CPI.
- **Microarchitectural Convergence:** Modern x86 processors crack CISC instructions into internal RISC μops, validating the core RISC execution model.
- **Evaluation Metrics:** True architectural efficiency is measured via the Iron Law ($T_{\text{CPU}} = \frac{IC \times CPI}{f_{\text{clk}}}$) and Performance-per-Watt.

Self-Test Questions: ISA Principles & Mechanics

1. **Load/Store Boundary:** Why does a strict load/store architecture simplify pipelined execution compared to an architecture permitting memory operands in ALU instructions?
2. **Parallel Decode Efficiency:** Why is an 8-wide instruction decoder significantly simpler and more power-efficient for a fixed 32-bit RISC ISA than for variable-length CISC?
3. **Micro-operation Translation:** When an x86 processor executes `add [rdi + 8], rax`, how does the front-end decoder translate this instruction into internal μops?

Self-Test Questions: Quantitative Analysis & Power

1. **Iron Law Calculation:** A RISC core executes 1.5×10^9 instructions at CPI $= 1.1$ on a 3.0 GHz CPU. A CISC core executes 1.1×10^9 instructions at CPI $= 1.6$ on a 3.2 GHz CPU. Which finishes faster?
2. **Front-End Power Cost:** Why does instruction decoding consume a higher percentage of core power in modern x86 processors than in AArch64 cores?
3. **Performance-per-Watt:** Why has Performance-per-Watt superseded raw clock frequency as the primary design metric in modern computing?

Looking Ahead

Next Lecture Preview

Our Next Topic: The Hardware/Software Interface & AArch64 Programmer's Model

- **Architectural State:** Defining the precise software-visible hardware contract (registers, condition flags, stack pointer, and memory space).
- **AArch64 Register File:** Examining register naming conventions (`x0`–`x30` vs. `w0`–`w30`), the zero register (`xzr`), and standard ABI calling conventions.
- **Instruction Mechanics:** Exploring foundational instruction classes, memory addressing modes, and control flow translation.